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Forecasting electoral violence☆

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ABSTRACT

Electoral violence remains a significant challenge worldwide. It not only threatens to undermine the legitimacy and fairness of electoral outcomes, but often has serious repercussions on political stability broadly. The ability to prevent electoral violence is critical for safeguarding democracy and ensuring peaceful transitions of political power. Predicting which elections are at risk of violence is a crucial step in effective prevention. In this study, we develop a set of machine-learning models to forecast the likelihood of electoral violence worldwide. Using diverse data sources, which include economic indicators, the history of electoral violence, political instability, and digital vulnerability, we predict the risk of electoral violence on a scale ranging from no violence to severe violence. Our final forecasts are produced by combining constituent models into an ensemble using a genetic algorithm. Out-of-sample evaluation of the system shows that the final model accurately distinguishes between different levels of risk. After validating our system on historical data, we generate out-of-sample probabilistic forecasts for national-level elections in 2025 and 2026. This research contributes to the field of political violence prediction by providing a medium-term data-driven forecasting tool for electoral violence.

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1. Introduction

A vast majority of countries in the world today hold elections in which citizens vote to fill the highest political offices of the state. Where elections are free and fair, they represent a cornerstone of democratic governance, providing a peaceful mechanism for transferring power and holding governments accountable.[1] Yet, electoral violence, i.e., violence seeking to influence the process or outcome of elections purposefully, represents a serious challenge to the integrity and legitimacy of electoral processes around the world (Birch et al., 2020). One in five national elections since 1946 has experienced significant levels of intimidation, harassment, and physical

☆ The numerical results presented in this manuscript were reproduced by the Editor-in-Chief on 7 September 2025.

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[1] In many non-democratic settings, elections are also held to identify opposition strongholds, signal the strength of the incumbent, provide trappings of democracy, and "divide and rule" tactics (Gandhi & Lust-Okar, 2009).

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violence, often leading to civilian fatalities (Hyde & Marinov, 2012a). In addition to the human suffering associated with all types of political violence, electoral violence risks undermining the legitimacy and fairness of the democratic process by influencing who stands for political office, who votes and whom they vote for, how votes are counted, and how electoral outcomes are enforced. The potentially far-reaching implications for political stability and social cohesion may even trigger broader political turmoil, including civil wars (Birch et al., 2020; Birch & Muchlinski, 2018). Whereas the holding of elections in authoritarian and hybrid regimes has aggravated the challenge, the problem is not restricted to the Global South or to weakly consolidated democracies, as recent episodes of electoral violence in the United States, Brazil, Turkey, and Hungary remind us. Thus, the promotion of peaceful elections is high on the political agenda for domestic and international agencies alike (Birch & Muchlinski, 2018; Kleinfeld & Sedaca, 2024).

More knowledge about when and where elections turn violent is in high demand across academic and policy-practitioner circles. This study responds to this demand. We build on recent methodological advances in the forecasting literature, along with the burgeoning literature on the causes of electoral violence, to predict the likelihood of violence in upcoming elections on a global scale. In recent years, forecasting has gained traction as a complementary approach to understanding political violence and assessing its consequences (e.g. Butcher et al., 2020; Hegre et al., 2019, 2021; Vesco et al., 2022). Prominent forecasting projects have focused on predicting various forms of political violence, such as the likelihood of future civil wars, fatalities in ongoing wars, and the likelihood of mass atrocities (for a review, see Rød et al. (2024)). However, research on electoral violence has, thus far, primarily been studied through an explanatory framework aimed at understanding the circumstances when it occurs and its consequences for democratic processes (e.g. Fjelde & Höglund, 2014-12; Hafner-Burton et al., 2013), rather than through a predictive framework that allows policymakers and stakeholders to take proactive steps to minimize its risk and consequences. This study aims to fill this gap by introducing a forecasting system that predicts electoral violence globally. The resulting forecasts can then be used to inform policy decisions and preventive measures aimed at reducing the risk of electoral violence in the future.

Our prediction system utilizes random forest classifiers trained on historical data on electoral violence, alongside a comprehensive set of structural, political, and socioeconomic factors, to predict electoral violence on a three-level ordinal scale: no violence, moderate violence, and severe violence. Our predictor features are grouped into thematic constituent models, which are combined for the final forecast using a genetic algorithm. Evaluating our final, weighted prediction models on historical data from 2015 to 2024 reveals that our model can accurately predict the level of electoral violence one and two calendar years ahead of elections in 417 and 421 out of 507 elections, respectively. The models also perform well on other important evaluation metrics such as the Brier score and the AUPR and AUROC metrics, and outperform the baseline models on all primary metrics.

Having validated the model's predictive performance on recent historical data, we then use the full dataset through the end of 2024 to generate forward-looking, probabilistic forecasts of electoral violence for upcoming elections in 2025 and 2026. These forecasts indicate an elevated risk of electoral violence, primarily in countries with a history of electoral violence and those with a mixed democratic record, as well as in consolidated democracies such as the United States. By publishing forward-looking predictions on a global scale, this paper makes a novel contribution to the study of electoral violence. It demonstrates how forecasting tools can be used to inform early warning efforts and guide preventive action in advance of elections.

2. Electoral violence and its determinants

Electoral violence is a complex phenomenon that can take many forms and be employed by various actors at different stages of the electoral process. Broadly defined, electoral violence refers to any use of violence, coercion, or intimidation by any political actor aimed at influencing the outcome of an election or undermining the credibility of the electoral process (for conceptual discussions, see for example Birch et al., 2020; Höglund, 2009). Electoral violence can take many forms, including lethal (e.g., assassinations, terrorist attacks) and non-lethal physical violence (e.g., unlawful arrests, beatings, assaults), as well as forms of intimidation and threats that may not involve physical harm but aim to influence the behavior of voters, candidates, or election officials. It can occur throughout the electoral cycle, from voter registration and party primaries (e.g. Seeberg et al., 2018) to the implementation of results. Research suggests it often manifests differently, depending on the stage of the electoral process (Straus & Taylor, 2009). The pre-electoral period frequently sees violence aimed at shaping whether citizens vote and who they vote for, for example, through voter suppression, intimidation of opposition candidates, and crackdown on political rallies. On election day, violence is used to disrupt voting or directly manipulate the ballot, e.g., through attacks against polling stations or electoral workers. The post-electoral period is often characterized by violence aimed at contesting the results of the election and frequently manifests as violent protests or riots against electoral fraud, and regime crackdown on the same.

Over the past 15 years, a substantial body of literature has emerged, seeking to understand the characteristics of election violence and its underlying drivers. A starting point for much of this literature is the observation that the introduction of democratic processes, such as competitive elections and broad participation, oftentimes is associated with an increased risk of political violence, both among countries transitioning from authoritarian rule or emerging from civil war (e.g. Brancati & Snyder, 2013; Cederman et al., 2010; Hegre et al., 2001). Research has also shown how the risk of political violence is significantly higher during elections compared to other periods (Harish & Little, 2017).

Research that explicitly attempts to understand why some elections turn violent and others do not has focused on the institutional framework. Salehyan and Linebarger (2014) find that the risk of election violence is particularly high in hybrid regimes, i.e., regimes that combine autocratic and democratic features. Generally, political institutions that diffuse and share power reduce the risk of violence. Hafner-Burton et al. (2013), for example, identify strong constraints on the executive as an important inhibitor for government-perpetrated violence. Fjelde and Höglund (2016) find that majoritarian electoral institutions, which heighten the stakes of the electoral contest, are associated with a higher risk of election violence. Different types of elections see different risks of violence (Daxecker & Rauschenbach, 2019). More generally, electoral violence becomes more likely where formal institutions are weak, a finding that relates both to actors that compete in elections such as political parties, (Fjelde, 2020), as well as to the state as a whole (Birch et al., 2020).

Where formal institutions are weak, informal institutions such as clientelism and corruption prevail, which, with their divisive and exclusionary logic, often precipitate violence, for example, through competition over economic resources like land (Berenschot, 2020; Boone, 2011; Klaus, 2020; Klaus & Mitchell, 2015). These studies also highlight how institutional frameworks work in tandem with societal features, such as land tenure or inequalities, to increase the risk of violent elections. Majoritarian, "winner-takes-all" political frameworks, for example, are particularly conflict-prone in societies where large ethnopolitical groups are excluded from power and where economic disparities are high (Fjelde & Höglund, 2016).

Other scholars have focused more specifically on the conduct of the elections. Electoral management bodies, in particular, have a central role in ensuring high-integrity and peaceful elections (Opitz et al., 2013; Ruiz-Rufino & Birch, 2020). Some studies have also focused on more immediate features of the electoral contest, highlighting features that increase uncertainty about the electoral outcome, such as whether the incumbent stands for reelection (Straus & Taylor, 2009) or electoral competitiveness (Daxecker, 2020; Wilkinson, 2004). Patterns of competitiveness and stronghold can also explain the spatial distribution of violence within a country (Daxecker et al., 2024; Wahman & Goldring, 2020). Third-party actors also matter, but the identified impacts are somewhat ambiguous. Fjelde and Smidt (2022) find that UN peacekeepers reduce the risk of election violence by non-state actors but are less efficient when it comes to halting government-perpetrated abuse in post-conflict elections. Moreover, whereas international observers contribute to peaceful elections by detecting and deterring fraud and overt (including violent) forms of electoral manipulation on election day, their presence might temporally and geographically shift violence to less observed periods and geographical areas (Daxecker, 2014). By exposing cheating and irregularities, international observers might also heighten the risk of post-election mobilization to protest fraudulent electoral outcomes (Borzyskowski, 2013, 2019; Daxecker, 2012).

3. Constructing a prediction target for electoral violence

Understanding the multifaceted nature of electoral violence is crucial for developing effective forecasts. To build a forecasting system capable of making such predictions, we must also construct a prediction target that reflects the complexity of the phenomenon while remaining tractable for empirical modeling.

Several projects exist that code electoral violence, such as the Countries at Risk of Electoral Violence (CREV) project (Birch & Muchlinski, 2020), the Deadly Electoral Conflicts dataset (DECO) (Fjelde & Höglund, 2022), the National Elections Across Democracy and Autocracy (NELDA) project (Hyde & Marinov, 2012b), and the Varieties of Democracy (V-DEM) project (Coppedge et al., 2024). These projects code electoral violence in different ways, focusing on different aspects of the phenomenon, and have different scopes in terms of time and space. Our goal in this paper is to produce global-scale forecasts of electoral violence. As electoral violence is a complex and multifaceted phenomenon, we believe that the prediction target should capture the full range of electoral violence, including different types of actors, forms of violence, and stages of the electoral process. As such, we need to choose a prediction target that fulfills three criteria. First, the data source must have global coverage of elections, encompassing both established and emerging democracies, as well as elections in countries not classified as democracies. Second, the data source must cover a sufficiently long time period to enable the training of machine learning models. Third, the data source must code electoral violence in a manner that captures the full range of violence, encompassing various types of actors, forms of violence, and stages of the electoral process.

To this end, we have chosen to work with two indicators from the V-DEM project (Coppedge et al., 2024; Pemstein et al., 2024). The first indicator is the *Election Government Intimidation* (v2elintim), measuring the extent to which the government uses intimidation and harassment to influence the outcome of elections. The second indicator is the *Election other electoral violence* (v2elpeace), measuring the extent to which actors other than the government use violence and coercion to influence the outcome of elections (Coppedge et al., 2024). Both of these indicators are coded by country-level experts on a five-point scale, ranging from 0 (severe violence) to 4 (no violence), and cover the campaign period, election day, and post-election process for each national-level election or group of elections.[2] As multiple experts can code each election, the value of the indicator can take any value between 0 and 4 (Coppedge et al., 2024; Pemstein et al., 2024). The full definitions of the indicators can be found in the V-DEM codebook (Coppedge et al., 2024).

To create a single indicator of the level of electoral violence to be used as the prediction target, we take the minimum, i.e. the indicator with the highest level

[2] e.g. if a country has both presidential and legislative elections on the same day, these are coded as one group of elections

of violence, of the two indicators for each election and re-code this to a three point-ordinal scale: *no electoral violence*, for elections where the minimum value is 3 or above, *moderate electoral violence*, for elections where the minimum value is 1.5 or above, and *severe electoral violence*, for elections where the minimum value is below 1.5. The thresholds were selected to roughly correspond to the levels of severity in the original indicators, where values of 0 and 1 correspond to the most severe violence, values of 2 and 3 correspond to moderate violence, and a value of 4 corresponds to no violence. Choosing the minimum, rather than the mean, of the two indicators allows us to classify cases where either the government or other actors use violence as severe, even if the other actor does not use violence. This allows us to capture the full range of electoral violence, from no violence to severe violence, in a single indicator. The threshold for *no electoral violence* was set to 3 or above rather than 3.5 to not classify borderline cases with minimal levels of violence as *moderate electoral violence*.[3]

As the forecasting system aims to output forecasts for the annual risk of electoral violence, we aggregate the data to the country-year level, taking the maximum level of electoral violence observed in each country-year with an election. This means that if a country has multiple elections in a single year, the prediction target for that year will be the maximum level of violence observed across these elections.[4] Furthermore, we only include country-years with at least one national-level election, such as a presidential election or an election to the legislative assembly, and exclude country-years that only feature regional elections, referendums, and/or international-level elections (e.g., elections to the European parliament). This is done to ensure that the prediction target is comparable across countries and contexts, and to avoid conflating distinct political processes that may have different dynamics in terms of the drivers and manifestations of electoral violence. Lastly, we only include elections from 1989 onwards. Using 1989 as a starting point is motivated by the fact that the post-Cold War era has seen a significant increase in the number of elections held globally, and that the dynamics of electoral violence may have changed in this period compared to earlier periods.[5]

3.1. Types of electoral violence in the data

Based on the definitions above, we were able to extract data on the electoral violence in 1794 *country-election-years* across 170 countries between 1989 and 2024. This includes 823 country-election-years without electoral violence, 690 with moderate violence, and 281 with severe violence. The data set comprises elections in 170 countries. The level of coded electoral violence in the last observed election for all countries in our data set is shown in Fig. 1 below. The figure shows that the level of electoral violence varies significantly across countries, with some experiencing severe violence in the most recent observed election. In contrast, others have not experienced any violence in the same period.

To illustrate how different our indicator captures forms of electoral violence, we briefly highlight examples from the last 15 years corresponding to the severe and moderate categories on our three-point ordinal scale.

Severe electoral violence includes cases where either the government or non-government actors (or both) engage in high-intensity violence. Notable examples include the 2024 presidential election in Venezuela and the 2015 Burundian election, both marked by widespread repression and violent contestation involving both actor types. Other cases, such as post–Arab Spring elections in Egypt (2014–2023), Russia (2024), and Belarus (2020–2024), feature state-led violence without corresponding non-governmental action. In contrast, Afghanistan (2018–2019) and the Ivory Coast (2010–2011, 2020–2021) exemplify non-state-driven violence, including insurgent attacks and opposition-led unrest, with more restrained state responses.

Moderate electoral violence is over twice as common as severe violence and typically involves lower-intensity or localized incidents. Elections in countries like Turkey, Bangladesh, and Kenya are typical examples of such violence, which often reflect persistent, low-intensity electoral contestation, typically involving localized clashes, partisan militias, and institutional harassment without tipping into extreme and widespread electoral violence. Other cases, like Russia (before the 2024 election), Kazakhstan, and Iran (2012–2020), reflect moderate government-led repression through repression of protests, legal coercion, and arrests. Non-governmental actors are the main perpetrators in countries such as the United States (2020 and 2024), with violence surrounding political rallies, election administration, and post-election protests, and elections in countries such as Colombia (2010–2022), Mexico (2006–2021), and Guatemala (2011–2023), where criminal organizations, partisan groups, or local actors have employed intimidation and violence to influence outcomes or retaliate against political opponents.

4. Methods

Our forecasting system is based on a set of thematic constituent models, each designed to capture distinct sets of features relevant to predicting electoral violence. A subset of these models is then combined, using a genetic algorithm, into a weighted ensemble model that produces the final forecast of electoral violence. This approach aligns with the current state of the art in conflict forecasting (Hegre et al., 2019, 2021). In total, we tested 33 different constituent models containing a variety of features, including the history of electoral violence,

[3] A total of 192 country-election-years in the training data have the minimum value across the two indicators in the range of 3 to 3.5. As a robustness check, we also tested the prediction system with a threshold of 3.5 for *no electoral violence*, which yielded similar performance to the model with a threshold of 3, but naturally with fewer cases classified as *no electoral violence*.

[4] A manual review of the data showed that the level of electoral violence coded was the same for all country-years with multiple elections in the same year.

[5] Experiments with different starting points for the data, such as 1975 and 1960, did not yield any meaningful differences in the performance of the prediction system or the predicted probabilities of electoral violence in the test data.

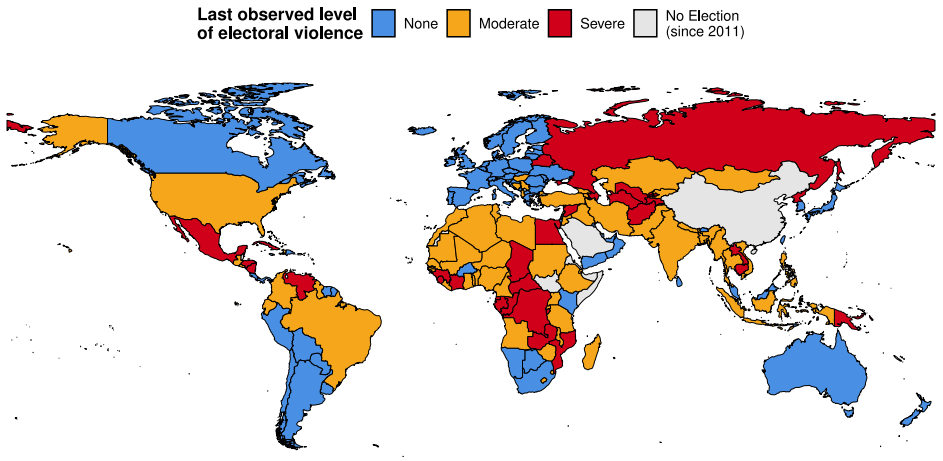


Fig. 1. Map of coded electoral violence during the last observed national election. Territories not coded as independent by VDEM and countries with no election in the last 15 years are excluded.

electoral characteristics, and a wide range of other political, economic, social, and geographic variables. The features are drawn from the Varieties of Democracy (V-DEM) project (Coppedge et al., 2024; Pemstein et al., 2024), the World Bank's World Development Indicators (WDI) (WorldBank, 2023), and the Digital Society Project (Mechkova et al., 2024). While the constituent models are organized thematically, we have taken an exploratory approach to their exact composition, with several models containing overlapping features, as well as some models that use narrower or broader sets of variables within a theme, and others that span multiple thematic areas. Our expectation is not that all models will be useful for forecasting; however, including potentially uninformative models allows the ensemble method to perform implicit model selection, ensuring that we do not overlook configurations that may capture distinct or complementary signals. A description of all 33 tested thematic constituent models, including the features included in each, is available in Appendix A1.

As the goal of the forecasting system is to provide global medium-term forecasts of electoral violence up to two calendar years into the future, using only data available at the end of the previous year, the features included must adhere to this constraint. This means that we do not include features such as campaign events, polling data, or other real-time data sources that would not be available at the end of the year but which are likely to be important for short-term forecasting. Instead, we focus on structural features and characteristics that measure the political and electoral system (including the history of electoral violence), which are generalizable across different cases and are available at the end of the year from the data sources mentioned above.

Predictions in the forecasting system were made as probabilities for the three different levels of electoral violence (no, moderate, and severe). As our prediction algorithm, we use a standard Random Forest classifier with probability estimates (Breiman, 2001). The Random Forest classifier is a machine learning method that fits a number of decision tree classifiers on random subsamples of the training data and uses averaging to improve predictive accuracy and control overfitting. The Random Forest classifier is commonly used for predicting political violence (Hegre et al., 2019, 2021; Muchlinski et al., 2016), and has been demonstrated to perform well in various contexts.

Unlike more complex machine learning models, such as deep learning models and gradient-boosted models, the Random Forest model is relatively robust to overfitting and does not require extensive hyperparameter tuning. This makes it a good choice for our forecasting system, as the available training data is relatively limited, which can pose a problem for hyperparameter tuning. Initial experiments using a gradient-boosted model showed a high degree of instability in the hyperparameters, making it difficult to obtain consistent results across different experimental runs. The gradient-boosted model also did not outperform the Random Forest model in terms of predictive performance, which further supported our decision to use the Random Forest model for our forecasting system.

4.1. Training and evaluation

To ensure a robust evaluation of the forecasting system, it is essential to assess performance on data not used during model training. This guards against overfitting, prevents data leakage, and helps evaluate how well the model generalizes to new data (Hernández-Orallo et al., 2012; Ying, 2019).

While the standard approach is to set aside a static holdout set, this can be data-intensive and inefficient when working with limited samples. Instead, we adopt a rolling test window strategy, a form of time-series cross-validation, which balances the need for out-of-sample evaluation with efficient use of available data. This approach also closely mirrors real-world forecasting, where models are re-trained as new data becomes available (Bergmeir & Benítez, 2012).

Specifically, we use a 10-year rolling test window covering the period from 2015 to 2024. The process begins with a model trained on data from 1989 to 2014, which

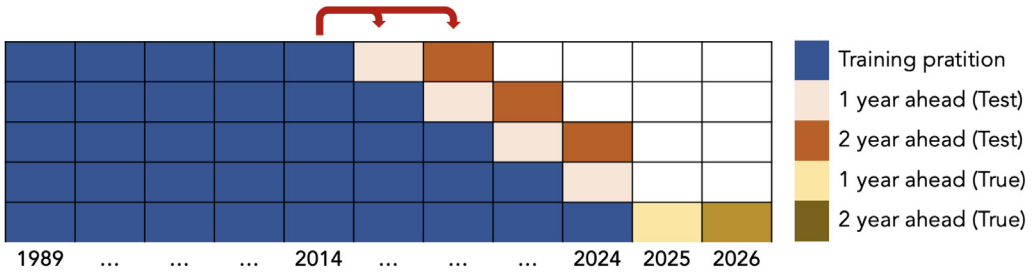


Fig. 2. Graphical illustration of the rolling window approach. The dark blue cells represent the training partition for each iteration. One-year-ahead forecasts (in light beige) and two-year-ahead forecasts (in dark orange) are generated for each evaluation window. After each step, the training window is expanded by one year. The final forecasts for 2025 and 2026, shown in yellow and brown, represent true out-of-sample predictions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

is used to generate one- and two-year-ahead forecasts for 2015 and 2016, respectively.[6] The training window is then updated to include data up to 2015, and forecasts are produced for 2016 and 2017. This procedure is repeated iteratively until 2024, at which point we generate true out-of-sample forecasts for 2025 and 2026.

This rolling-window framework enables us to maximize the use of historical data while maintaining a strict separation between training and evaluation periods, thereby mimicking a real-world forecasting scenario. We train and evaluate the models for the one-year and two-year forecasting horizons separately, as the performance of the models may vary across the different horizons. The full process is illustrated in Fig. 2.

4.2. Performance metrics

To evaluate the performance of the models in our rolling test window, we use four different performance metrics: accuracy, area under the receiver operating characteristic curve (AUROC), area under the precision–recall curve (AUPR), and the Brier score.

Accuracy

Accuracy measures the proportion of correctly classified instances out of all instances in the test set. Accuracy is an intuitive metric that is easy to interpret, but can be misleading when classes are imbalanced, when costs of different types of errors are not equal, or when the difficulty of the classification task varies across different classes.

AUROC

The area under the receiver operating characteristic curve (AUROC) measures the trade-off between the true positive rate, i.e., the proportion of true positives classified as positives, and the false positive rate, i.e., the proportion of true negatives classified as positives, across different thresholds for classifying instances. The AUROC ranges from 0 to 1, where a value of 0.5 indicates that the model performs no better than random, and a value of 1 indicates perfect performance. The AUROC is a useful metric for evaluating the overall performance of a classification model, but can be misleading when the classes are imbalanced (Hernández-Orallo et al., 2012).

While the AUROC metric was initially designed for binary classification, it can be adapted for multi-class problems using the "one-vs-rest" approach.[7] Here, each class is treated individually as the "positive" class, with the others as "negative." We then average these scores to get the overall AUROC, allowing us to effectively evaluate our multi-class forecasting system (Hernández-Orallo et al., 2012).

AUPR

The area under the precision–recall curve (AUPR), also known as average precision, measures the trade-off between precision, i.e., the proportion of true positives among all instances classified as positives, and the true positive rate (recall) across different thresholds for classifying instances. The AUPR ranges from 0 to 1, where a value of 0 indicates that the model performs no better than random, and a value of 1 indicates perfect performance. The AUPR is a useful metric for evaluating the performance of a classification model when classes are imbalanced, as it focuses on the positive class (Hernández-Orallo et al., 2012).

Similar to the AUROC metric, the AUPR metric was designed for binary classification problems but can be extended to multi-class classification problems using the one-vs-rest approach.

Brier score

The Brier score is a proper scoring rule that measures the mean squared difference between the predicted probabilities and the actual outcomes. The Brier score ranges from 0 to 1, where a value of 0 indicates perfect performance. The Brier score is a useful metric for evaluating the calibration of a classification model, as it measures the accuracy of the predicted probabilities. The Brier score is particularly useful when the predicted probabilities are used to make decisions, as it directly measures the quality of the predictions (Hernández-Orallo et al., 2012).

[6] To ensure that all evaluations are based on comparable election years, we include an additional model trained on data up to 2013 for the two-year-ahead forecast for 2015.

[7] Other alternatives, such as class-weighted average AUROC, can also be computed.

4.3. Ensembling by genetic algorithm

To improve the forecasting performance of our system, we combine the predictions of multiple constituent models using an ensemble approach. Ensembling is a well-established strategy for increasing robustness, reducing overfitting, and improving generalization in machine learning systems (see for instance Hegre et al., 2019; Montgomery et al., 2012; Rød et al., 2024). There are several methods for creating ensembles, including naive ensembles, Bayesian model averaging, bagging, boosting, and stacking. In this paper, we use a genetic algorithm to optimize the weights of the constituent models in the ensemble (Holland, 1992; Sivanandam et al., 2008).

The genetic algorithm was configured to assign weights to the predictions of the constituent models in the ensemble and calculate a weighted average of them as the final ensemble prediction. It begins with 1,000 randomly initialized individuals (weight vectors for the predictions of the constituent models) and evolves over 200 generations. In each generation, the algorithm evaluates the fitness of each individual based on the ensemble's performance on all out-of-sample predictions available up to that year in the training data. The fitness function is defined as the inverse of the Brier score, such that individuals with lower Brier scores have higher fitness. Based on the fitness of each individual, the algorithm selects a number of the best individuals ("elitism"). It generates new individuals through crossover ("mating") and mutation, as well as replacement of the worst individuals. We apply standard genetic operations with an elitism rate of 0.01, a crossover rate of 0.89, and a kill/replacement rate of 0.1.

To promote diversity and avoid premature convergence ("inbreeding"), we implement two mechanisms: (1) a *relative similarity threshold* for mating, which prevents mating between individuals whose cosine similarity exceeds the 75th percentile in the population; and (2) a *similarity-weighted dual mutation scheme*, where mutation and replacement rates are scaled by each individual's similarity to the median similarity of individuals in the population. This dual mutation scheme was set up to have a gene-mutation rate of 0.1 and a gene-replacement rate of 0.01 as a baseline, and then scaled by the relative similarity of the individual to the population, such that individuals with higher similarity to the population compared to the median had a higher mutation rate.

We further apply *sparsity-inducing regularization* such that weights below 0.02 were iteratively set to zero and redistributed among the remaining non-zero weights to force sparsity and model selection in the ensemble. This encourages model selection by forcing the algorithm to prioritize among highly correlated models.

The genetic algorithm was trained separately for the one- and two-year-ahead forecasts and was re-trained for each year in the rolling test window using the out-of-sample predictions of the constituent models up to that year as input. To ensure a sufficient number of out-of-sample predictions for the genetic algorithm, we extended the rolling test window for training the genetic algorithm by five years, so that the genetic algorithm was trained on the out-of-sample predictions from 2010 to the last year of the training data.[8]

4.3.1. Ensembles for the final forecasts

Since the genetic algorithm is retrained annually, the ensemble weights vary from year to year based on the recent performance of constituent models. This dynamic weighting reflects the fact that many constituent models are highly correlated, especially those that include features such as past electoral violence or irregularities.[9] Because of these high correlations, small differences in recent performance can lead to substantial shifts in assigned weights, especially under the enforced sparsity threshold of 0.02, which in essence forces the genetic algorithm to choose between models when they are highly correlated.

The final weights for each forecast year, including the out-of-sample forecasts for 2025 and 2026, are presented in Tables B3 and B4 in Appendix B. Across years, the algorithm selects between 7 and 12 models with non-zero weights. While weight allocations vary, several general patterns emerge:

- The constituent model focusing on the *history of electoral violence* model receives the overall highest average weight (33% for one-year ahead, 20% for two-year ahead).
- The five models focused on *electoral irregularities and electoral characteristics* collectively contribute 8% and 23% on average.
- The most *broad-based models*, those incorporating multiple themes or drawing on the full V-Dem and V-Dem + WDI datasets, also play a key role (31% and 23%, respectively).
- Models based on the *Digital Society Project* contribute a modest share, averaging 9% and 7%.
- Several models, most notably those only including WDI data and a few of the narrower VDEM (e.g., the gender, civil liberties, and accountability models) and DSP models, receive no or very low weights in most years.

The high weights assigned to models focused on electoral violence, irregularities, and characteristics are unsurprising, given the well-documented persistence of violence across elections. At the same time, the ensemble consistently incorporates a diverse set of models, broadening its scope and improving its ability to capture the multifaceted nature of electoral violence. Furthermore, there appear to be some time trends in the weights of the models, where the weight of the history of electoral violence constituent model seems to be declining over

[8] i.e., such that the one-year-ahead forecast for 2015, the genetic algorithm is optimized using the out-of-sample predictions from 2010 to 2014, and for the two-year-ahead forecast for 2025, the genetic algorithm is optimized using the out-of-sample predictions from 2010 to 2023.

[9] In the one-year-ahead window, the minimum correlation of predictions among the top 10 constituent models (by Brier score) is 0.92 for no violence, 0.87 for moderate violence, and 0.85 for severe violence.

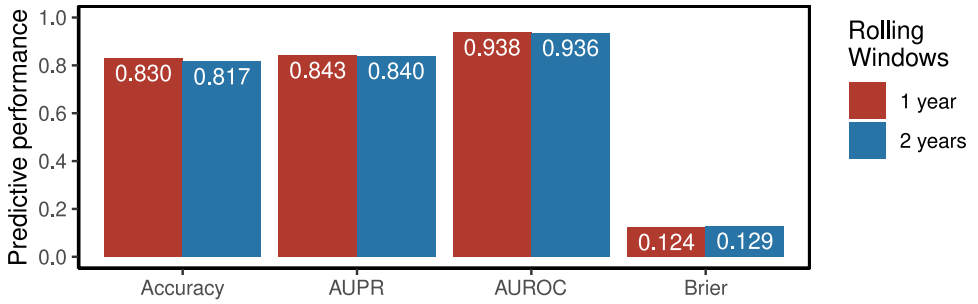


Fig. 3. Evaluation metrics: Accuracy, AUPR, and AUROC for 507 country years with elections across the 2015–2024 period, for models trained on data up to one year before the evaluation year (1-year rolling window) or two years prior (2-year rolling window).

time in favor of models focused on electoral irregularities from the last election. These trends strengthen the case for retraining the genetic algorithm annually in the system.

4.4. Benchmark models

To evaluate the performance of both the constituent models and the genetically weighted ensemble, we compare them to a set of benchmark models. First, we include two simple benchmarks: (1) a *persistence* model, which predicts the same outcome as the previous election, and (2) a *climatology-inspired* model, which assigns the probability of electoral violence based on the observed proportion of different violence levels in the last five country-wide elections.[10] While simple, persistence and climatology models are commonly used benchmarks in both meteorology, conflict research, and other fields (e.g., Hegre et al., 2024; Jung & Broadwater, 2014). Second, as a benchmark specific to the genetically weighted ensemble, we include a *naive ensemble* model, which generates predictions by taking the simple average across all constituent models.

5. Results

5.1. Out-of-sample evaluation

The predictive performance of the genetically weighted ensemble in the rolling test window from 2015 to 2024 is shown in Fig. 3. The ensemble models show high performance across all evaluation metrics, with accuracy levels of $\geq$ 80% and AUPR $\geq$ 0.8 and AUROC $\geq$ 0.9, indicating that they perform very well in distinguishing between positive and negative classes across different classification thresholds. Additionally, Brier scores around 0.1 are also indicative of well-calibrated predictions, highlighting that the models not only distinguish between classes well but also do so with high sharpness.

The corresponding evaluation metrics for all 33 thematic constituent models and the three benchmark models can be found in Tables B1 and B2 in Appendix B. Comparing the performance of the ensemble models to the constituent models, the genetically weighted ensemble outperforms all constituent models in both the one- and two year-ahead forecasts on the Brier score. It also outperforms all constituent models on the AUROC and AUPR metrics for both forecast horizons. While the performance differences among the top thematic models are relatively small, especially those focused on the history of electoral violence and election characteristics, the ensemble approach is more robust and generalizable, as it integrates complementary strengths across models and is less prone to overfitting.

Comparing the performance of the genetically weighted ensemble to that of the benchmark models provides further insight into performance. The *persistence* and *climatology-inspired* models yield relatively high accuracy given their simplicity (77%–78% and 72%–73%, respectively, in the two forecasting windows), but perform poorly in terms of calibration, as reflected by their high Brier scores. Consequently, they rank among the lowest-performing models when evaluated on the Brier score. The *naive ensemble* performs well across all metrics and achieves competitive results, but does not outperform the best individual constituent models and is ranked 11th and 10th overall on the Brier score in the two forecasting windows. The naive ensemble, however, yields marginally higher AUPR and AUROC scores in both forecasting windows. Given that the genetic algorithm is optimized for the Brier score rather than the AUPR or AUROC, this is not an entirely unexpected result.

Overall, the comparison with the constituent models and the benchmark models validates the effectiveness of the genetically weighted ensemble approach. Furthermore, the genetically weighted ensemble also enables inferences about the relative importance of constituent models, a feature that the naive ensemble cannot provide, making it not only more accurate but also more informative for interpretation and future refinement.

Fig. 4 shows the observed level of electoral violence against the predicted level across the 507 country-years with elections over the 2015–2024 period.[11] One year

[10] The choice of five elections is arbitrary but broadly reflects the average level of electoral violence over the past 10–20 years, depending on the country's electoral cycle. This provides a reasonable estimate of historical violence levels without being overly sensitive to either the most recent or the most distant elections. Tests with alternative numbers of elections yielded similar levels of predictive performance.

[11] The 'Predicted level' is taken as the level of violence with the highest predicted probability from the genetically weighted ensemble.

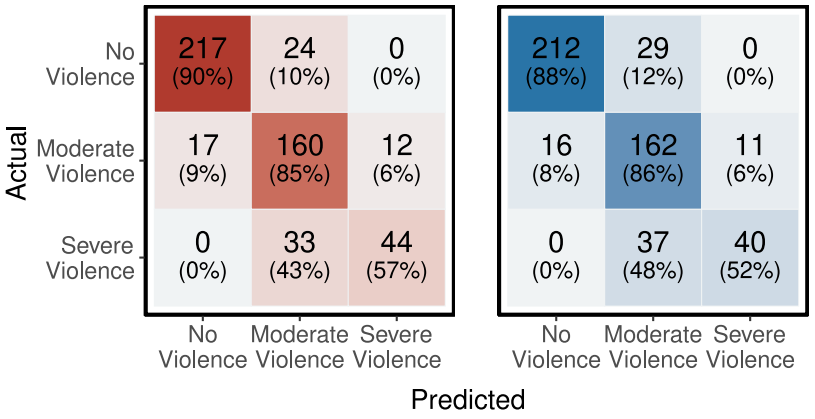


Fig. 4. Confusion matrices: Actual outcome versus predicted outcome for 507 country years with elections across the 2015–24 period, for models trained on data up to one year before the evaluation year (1-year rolling window, left panel) or two years prior (2-year rolling window, right panel). Percentages are row-wise percentages.

into the future, the model predicted 56 cases with severe electoral violence. Of these 56, 44 did, in fact, experience severe violence, while the remaining 12 experienced a moderate level of violence. None of the 56 cases predicted to have severe violence experienced any electoral violence. Thus, there are only a few instances where the model exaggerates the risk of violence. The rate of false alarms is also low: one year into the future, the model predicted 234 out of the 507 country years to have *no violence*. Only 7% of these predictions were incorrect. Of the 77 elections that experienced severe violence in the period, the model correctly predicts 57% of cases. In no case where severe electoral violence occurred did the model predict that no violence would be the most likely outcome.

The results of the out-of-sample evaluation in the rolling test window show that the final ensemble models demonstrate strong predictive accuracy, effectively distinguishing between classes with high reliability and well-calibrated probabilities. They also outperform, or match, the individual thematic models, indicating that the predictions are robust. The low overall error rates and the zero misclassification between no and severe electoral violence further highlight the reliability of the model. This overall performance gives confidence in the model's ability to make accurate forecasts for the future.

5.2. Illustrative case examples, 2024

To further evaluate the predictions, we compare our forecasts with qualitative evidence from three selected electoral processes that took place in 2024. We focus on Venezuela, India, and the United States as three cases with different democratic traditions that have experienced varying levels and types of electoral violence in the past. Venezuela illustrates a case of a more autocratic regime holding contested elections with an organized opposition. India represents a multi-ethnic decentralized democracy where both regional and national dynamics influence electoral violence. The United States exemplifies a consolidated democracy where recent elections have nonetheless been marked by political polarization and growing concerns about electoral security.

5.2.1. Venezuela

The Venezuelan presidential election was held on July 27, 2024. Our model predicted a high likelihood of electoral violence, estimating a 55% probability of limited violence and a 38% probability of severe violence. The election was ultimately coded as involving severe levels of violence (see Fig. 1).

In the lead-up to the election, the government engaged in widespread repression. Opposition candidates were arbitrarily disqualified, including prominent figures such as the politician María Corina Machado, and activists were arrested or harassed. A month ahead of the election, at least 76 arbitrary detentions were documented during the campaign period (CEPAZ, 2024; Human Rights Watch, 2024b). The election result was widely contested, with both the opposition and the government claiming victory, leading to widespread protests and violence. The UN Human Rights Council's Fact-Finding Mission in Venezuela recorded at least 23 deaths between July 28 and August 8. Furthermore, over 1260 people were detained in connection with the election, with serious due process violations such as remote hearings, unjust charges, and restricted access to legal counsel (OHCHR, 2024).

5.2.2. India

The Indian general election took place between April 19th and June 1st, 2024. Our model predicted a high likelihood of electoral violence for this election, with an approximate 75% probability of limited violence and 14% probability of severe violence. In line with these predictions, the election was coded as showing moderate levels of violence (see Fig. 1).

The pre-election and election periods were marked by widespread violence and intimidation in several parts of India. This violence was both politically and communally charged, involving various actors, but mainly including supporters of the ruling Bharatiya Janata Party (BJP) and state agencies controlled by them, along with opposition parties. The electoral landscape in India has been shaped by physical altercations, targeted assassinations, and systemic suppression, particularly impacting minority groups

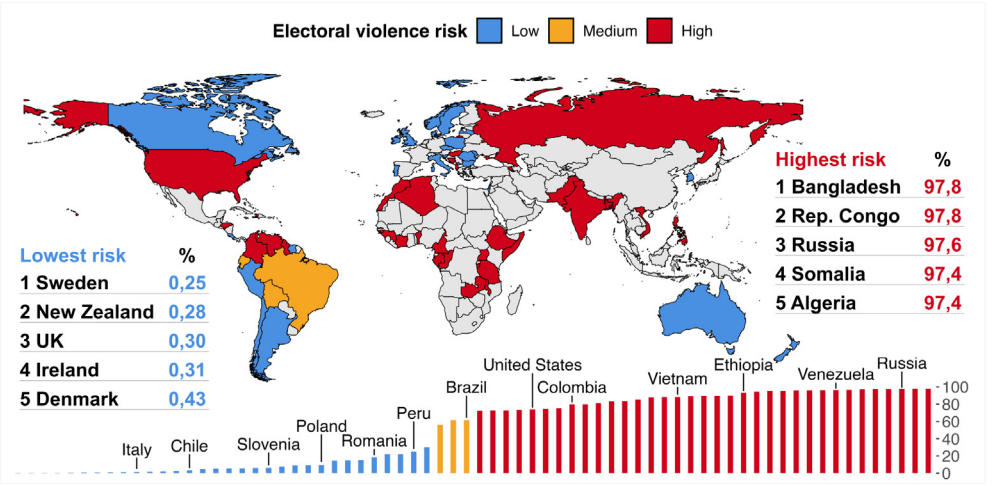


Fig. 5. Predicted probability for the risk of either moderate or severe election violence in 2025 or 2026. Countries without scheduled elections are marked with a gray shade. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

such as Muslims and Christians (Amnesty International, 2024; Human Rights Watch, 2024a)

One of the most notable examples comes from West Bengal, where tensions between the BJP and the Trinamool Congress (TMC) have resulted in repeated violent clashes. Violent encounters, including the use of crude bombs, stone-pelting, and car blockades, have been reported. Notably, in April 2024, more than 100 complaints of election-related violence were filed by both parties, illustrating the volatile atmosphere (The London Story, 2024).

5.2.3. United States

The United States presidential and general elections took place on November 5th, 2024. Our model predicts a relatively high likelihood of electoral violence for this election, with an approximate 51% probability of limited violence and 1% probability of severe violence. In line with these forecasts, the election was coded as involving moderate levels of violence.

The election cycle in the U.S. has been shaped by widespread intimidation and several significant instances of overt violence. Presidential candidates have been at the forefront of election-related violence and intimidation. Former President Donald Trump has faced two assassination attempts (Associated Press, 2024b). Another alleged attempt was thwarted in Palm Beach, Florida (Reuters, 2024a). Democratic candidates are also under threat. Vice President Kamala Harris and President Biden were targeted with violent online threats (Reuters, 2024b), further fuelling tensions. Election officials have also faced increased harassment and intimidation since the 2020 election, reflecting broader trends of rising threats against those tasked with upholding electoral integrity (Center, 2024). Recent threats include suspicious packages sent to election officials in 15 states, some containing hazardous substances such as fentanyl (Associated Press, 2024a).

5.3. Forecasts for 2025–2026

Using the genetically weighted ensemble, we forecast the probability of electoral violence for the years 2025 and 2026. The forecasts from our model are shown in Fig. 5 below, which displays the likelihood of *any* level of electoral violence, i.e., either moderate or severe electoral violence.[12] Countries with a low predicted risk of experiencing *any* level of electoral violence are shown in blue, countries at a medium predicted risk of experiencing *any* level of electoral violence are shown in yellow, and countries at a high risk of experiencing *any* level of electoral violence are shown in red.

The forecasts indicate that the likelihood of electoral violence is highest in countries with a history of violence or intimidation during elections, low levels of democracy, and a history of electoral violence, intimidation, and other irregularities. For example, the model predicts a high likelihood of electoral violence in countries such as Russia, the Republic of the Congo, and Venezuela. In contrast, consolidated democracies with little history of election irregularities, such as Denmark, Australia, and Ireland, are predicted to have a low likelihood of electoral violence. However, while both the highest predicted risk and the highest level of electoral violence intensity are forecasted in countries with low levels of democracy, this does not mean that this is a problem confined to non-consolidated democracies. Countries with long democratic traditions, including the United States, Hungary, and Colombia, also have elevated risks of experiencing any level of electoral violence. The United States, in particular, stands out, as the predicted probability of any electoral violence has risen from around 52% in the 2024 presidential election to above 70% for the midterm elections in 2026.

[12] Detailed predictions for all countries, regardless of whether they have scheduled elections in 2025 and 2026, can be found at the Kofi Annan Foundation's Electoral Vulnerability Index website or can be made available by the corresponding author upon request.

For these forecasts, it is important to note that the definition of electoral violence includes intimidation and harassment, which may not necessarily be expressed as overt, physical violence. Indeed, when repression is sufficiently severe, physical violence might not occur, even though the electoral environment is highly coercive. This caveat is important when interpreting the forecasts for countries with high levels of political repression.

6. Discussion

Forecasting of political violence has in recent years been propelled into the mainstream of conflict research, with a growing number of ambitious projects aiming to predict the likelihood of violence across various contexts (e.g. Hegre et al., 2019, 2021; Mueller & Rauh, 2018; Rød et al., 2024; Vesco et al., 2022). Predicting electoral violence is a particularly challenging task due to the complexity of the phenomenon and the multitude of factors that can influence its occurrence (Birch et al., 2020; Birch & Muchlinski, 2018; Höglund, 2009). Our forecasting system is built on thematic constituent models and ensembling, similar to state-of-the-art conflict forecasting systems (Hegre et al., 2019, 2021).

The evaluation of our model on historical data shows that our forecasting system can provide valuable insights into the likelihood of, and therefore prevention of, electoral violence. The genetically weighted ensemble performs particularly well on the performance metrics in out-of-sample evaluations from 2015 to 2024. This type of forward-looking prediction can be of great value to policymakers, election observers, and other stakeholders, as it helps identify countries at risk of violence and assists in implementing preventive measures to mitigate the risk. Importantly, comparisons with benchmark models underscore the added value of the weighted ensemble approach. While simple baselines, such as persistence and climatology-inspired models, achieve relatively high accuracy, their poor calibration, as reflected in their low Brier score performance, limits their practical utility for risk assessment.

In this paper, we use a broad definition of electoral violence, including intimidation and harassment as forms of violence. This is in line with the literature on electoral violence, which often includes a wide range of behaviors that can be seen as violent or coercive (Birch et al., 2020). However, it is essential to note that our forecasting system does not differentiate between various types of violence, only the level of violence. Because intimidation and harassment are included in our definition of electoral violence, highly repressive electoral environments where no physical violence is observed may still be classified as suffering from severe electoral violence. Therefore, countries with high levels of predicted risk of electoral violence may not necessarily experience widespread overt, physical violence. Notably, severe intimidation and coercion can be used in lieu of physical violence. However, even in highly repressive countries, a high predicted risk of electoral violence may still be a valuable signal for policymakers and other stakeholders, as it can help to identify countries where the risk of violence is particularly high, e.g., if the regime is challenged after the election, or if demonstrations or other forms of protest arise in conjunction with the election.

Our true out-of-sample forecasts for 2025 and 2026 identify countries of particular risk for electoral violence. The countries with the highest predicted risk and level of electoral violence are countries with a long history of electoral violence and that are not considered fully democratic. In these countries, the regime is likely to use violence to maintain power, either by subverting the electoral process using harassment and intimidation ahead of the election or through explicit physical violence following the election. Examples of countries in this category are Venezuela and Algeria. However, our forecasts also identify countries with a high level of predicted risk that have considerably longer democratic traditions, such as Ecuador, Brazil, and Hungary. In these countries, the risk is primarily for less severe levels of violence, but it still highlights the potential for violence even in consolidated democracies. Of particular note is also our prediction for electoral violence in the United States, which, according to our forecasting system, has a 74% likelihood of experiencing electoral violence in the 2026 midterm elections. This is the highest probability among all countries classified as liberal democracies by VDEM (Coppedge et al., 2024).

This study demonstrates that it is possible to build a forecasting system for electoral violence with a high degree of predictive accuracy using state-of-the-art machine learning tools. Forecasting political violence, however, is always fraught with ethical challenges, given the risk of self-fulfilling prophecies and the potential for misuse of the forecasts. However, we believe that by identifying countries at high risk, this forecasting system offers valuable insights for policymakers and election observers seeking to prevent violence and promote democratic integrity. Future work could expand the scope of these models to include more granular data and explore ways to integrate additional factors that may influence electoral violence. While challenges remain, particularly in distinguishing between types of violence, this study represents an important step toward more effective forecasting of electoral violence globally. Furthermore, the forecasting system presented here can serve as a useful baseline for future efforts to predict electoral violence, allowing researchers to build on the methods and data used in this study to refine and improve the accuracy of predictions.

CRediT authorship contribution statement

David Randahl: Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Maxine Leis:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Tim Gåsste:** Writing – review & editing, Writing – original draft, Validation, Methodology, Data curation, Conceptualization. **Hanne Fjelde:** Writing – review & editing, Validation, Conceptualization. **Håvard Hegre:** Writing – review & editing, Validation, Funding acquisition, Conceptualization. **Staffan I. Lindberg:** Writing – review & editing, Funding acquisition, Conceptualization. **Steven Wilson:** Writing – review & editing, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors utilized ChatGPT, Copilot, and Writeful to write code more efficiently, proofread, and correct grammatical errors and language flow in the article. After using these tools, the authors reviewed and edited the content as necessary, taking full responsibility for the publication's content.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: David Randahl reports a relationship with Kofi Annan Foundation that includes: consulting or advisory. Maxine Leis reports a relationship with Kofi Annan Foundation that includes: consulting or advisory. Tim Gasste reports a relationship with Kofi Annan Foundation that includes: consulting or advisory. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2025.09.003.

Data and code availability

The code and data needed to replicate the results of the study, along with a readme containing informmation of expected runtime and necessary hardware and enviroments are available at https://github.com/Doktorandahl/forecasting_ev_replication. The numerical results presented in this manuscript were reproduced by the Editor-in-Chief on 7 September 2025.

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